
Gluing Local Contexts into Global Meaning: A Sheaf-Theoretic Decomposition of Transformer Representations

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Abstract

1 A paraphrase pair is a small piece of evidence that two activations should agree.
2 We treat that as a sheaf condition. A cellular sheaf over a paraphrase graph defines
3 a coboundary δ^0 whose kernel holds activations that paste consistently across the
4 graph, and the rest of the Laplacian spectrum reads off how badly they fail to.
5 On an acyclic matching graph the construction collapses to projected within-pair
6 covariance, recovering a familiar quantity. Cycles change that. Stacking L layers
7 into a grid yields $b_1 = N(L - 1)$ algebraic 1-cycles, and a multi-paraphrase ring
8 per fact yields $b_1 = N$. The multi-layer grid produces a model-derived Hodge
9 harmonic mass that varies four orders of magnitude across nine architectures and
10 ranks the same models as their steering fragility (Spearman $\rho = 0.94$, $n = 6$,
11 exact permutation $p = 0.008$). The ring produces a cycle-aware H^0 that beats
12 pair-only H^0 on every one of eight architectures by a mean of +6.5 pp, a gain
13 within-pair covariance cannot reach. The single-layer construction is already strong:
14 across nine models from 124M to 13B parameters, the spectral complement of $L_{\mathcal{F}}$
15 carries 5.6 to 26.5 $\times$ the causal influence of variance-matched random subspaces
16 ($p < 10^{-15}$). On held-out CounterFact, sheaf H^0 at 20 dimensions beats LEACE’s
17 preserved subspace at full hidden dimension by a mean +17.1 pp, and swapping
18 the joint-PCA restriction map for random or identity projections barely moves
19 the number. The kernel selection drives the result, not the projection. On the
20 steering-fragile Llama-2-7B, contrastive sheaf steering preserves 31.0% of facts
21 against 4.2% for random perturbations ($n = 1000$, McNemar $p < 10^{-50}$). The
22 same operators recover collapse prevention: $\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$ is VICReg in
23 coordinates, holding encoder rank at 57–60/64 where naive consistency collapses
24 to 14–42/64.

25 1 Introduction

26 Transformer representations entangle what a sentence means with how it is phrased. The hidden
27 state for “The Eiffel Tower is in Paris” differs from “Paris is home to the Eiffel Tower” despite
28 identical semantic content. Existing tools decompose one activation into features [7, 11], test the
29 linear decodability of attributes already chosen [4], or relate two activation spaces to each other
30 [30, 38]. None of them measures cross-context consistency, the question of whether a feature encodes
31 the same information when phrasing changes and meaning stays fixed. The natural framing is
32 local-to-global. Each paraphrase pair gives one local witness that two activations should agree, and
33 many such witnesses must agree on a single global answer.

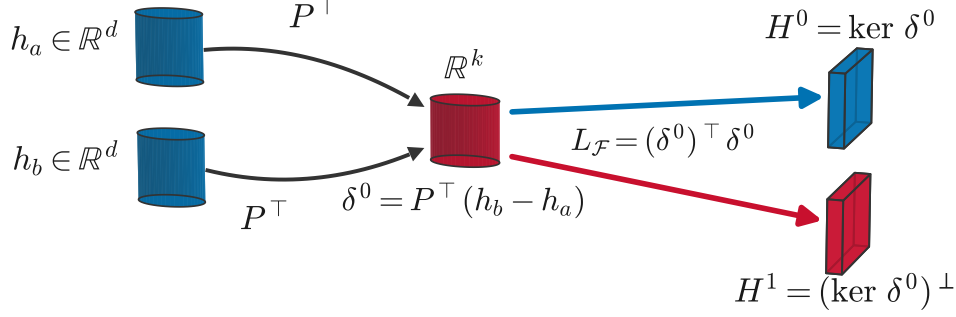


Figure 1: Sheaf decomposition of a paraphrase pair. Hidden states h_a, h_b pull back through $P^\top$ into edge space, the coboundary δ^0 measures pair disagreement, and the sheaf Laplacian $L_{\mathcal{F}} = (\delta^0)^\top \delta^0$ splits the space into $H^0 = \ker \delta^0$ (content stable across phrasing) and H^1 directions (context-dependent variation).

34 A cellular sheaf gives this construction a name. Nodes hold hidden states, edges connect paraphrase
 35 pairs, and a learned restriction map $P \in \mathbb{R}^{d \times k}$ projects each pair into a shared k -dimensional
 36 comparison space. The coboundary δ^0 measures pairwise disagreement, and the sheaf Laplacian
 37 $L_{\mathcal{F}} = (\delta^0)^\top \delta^0$ has a kernel $H^0 = \ker(\delta^0)$ of globally consistent sections together with a spectral
 38 complement of high-disagreement directions. We call the spectral complement “ H^1 directions” to
 39 indicate the role they play. Section 3 makes the construction precise, and Section 5 validates it
 40 functionally.

41 A single-layer matching graph has no cycles. Algebraic $H^1 = 0$, and the construction reduces to
 42 projected within-pair covariance. Stacking the same activations across layers brings cycles back.
 43 The grid has one node per (sentence, layer) pair, with horizontal edges between paraphrases in the
 44 same layer and vertical edges between adjacent layers along each sentence. It carries $b_1 = N(L - 1)$
 45 algebraic 1-cycles, one per elementary square, and each square asks whether the layer- l -to-layer- $l+1$
 46 transformation commutes with paraphrase equivalence. Across six architectures the model-derived
 47 Hodge harmonic mass at $L = 2$ varies by four orders of magnitude (10^{-4} to 0.08, Table 1) against
 48 a ~ 0.24 topological reference, and the ranking matches independently measured steering fragility.
 49 Cycles also arise without depth. Replacing the pair-only paraphrase forest with a multi-paraphrase
 50 ring per fact ($b_1 = N$) gives H^0 retrieval that beats pair-only H^0 by +6.5 pp on every one of eight
 51 architectures (Section 5.5). Both axes carry content that within-pair covariance does not.

52 Together these constructions give a cohomological view of the residual stream complementary to
 53 feature-level methods: local agreements between paraphrases glue into a global section through the
 54 kernel of the sheaf Laplacian, while phrasing-dependent variation lives in the spectral complement
 55 that no covariance proxy can recover.

56 Our contributions include:

- 57 1. We establish cellular sheaf cohomology as a transformer interpretability primitive at scale.
 58 Across nine architectures from 124M to 13B parameters, we validate the construction
 59 along three independent axes: functional (H^1 KL ratio over variance-matched controls),
 60 architectural (LEACE comparison on held-out CounterFact), and mechanistic (architecture-
 61 dependent steering fragility predicted by H^1 influence concentration).
- 62 2. Sheaf H^0 at 20 dimensions beats LEACE’s preserved subspace at full hidden dimension on
 63 held-out CounterFact retrieval across eight architectures (mean gap +17.9 pp, with 85.6%
 64 vs. 42.7% on GPT-2 and 87.1% vs. 77.7% on Mistral-7B), establishing the decomposition
 65 as a downstream-useful subspace method against a modern label-free baseline. On Llama-
 66 2-7B, contrastive sheaf steering preserves $7.4\times$ more facts than random (31.0% vs. 4.2%,
 67 $n=1000$).
- 68 3. Cycles in the paraphrase graph give downstream-useful H^0 , not just diagnostic harmonic
 69 mass. Multi-paraphrase rings ($b_1 = N$ per fact) beat the pair-only acyclic forest ($b_1 = 0$)
 70 on hard-CounterFact retrieval by +3 to +10 pp on every one of eight architectures (mean
 71 +6.5 pp, Appendix B.4). Multi-layer matching grids ($b_1 = N(L - 1)$) yield Hodge harmonic

mass spanning $[10^{-4}, 0.08]$ across six architectures against a ~ 0.24 topological reference, ranking the same models as their steering fragility.

4. A sheaf-regularized SSL objective $\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}(f)$ recasts an empirical fix as a coordinate of $\dim \ker(L_{\mathcal{F}})$. Its two terms (pair invariance and a kernel-dimension constraint) reproduce VICReg in coordinates, and across seven architectures the regulariser holds encoder rank at 57–60/64 where naive consistency collapses to 14–42/64.

2 Cellular Sheaves on Paraphrase Graphs

A cellular sheaf $\mathcal{F}$ on a graph $G = (V, E)$ assigns a vector space (stalk) $\mathcal{F}_v = \mathbb{R}^{d_v}$ to each node and a stalk $\mathcal{F}_e = \mathbb{R}^{d_e}$ to each edge, connected by linear restriction maps $\mathcal{F}_{v \sqsubseteq e} : \mathcal{F}_v \rightarrow \mathcal{F}_e$ at node-edge incidences [12, 24]. The coboundary operator $\delta^0 : C^0(\mathcal{F}) \rightarrow C^1(\mathcal{F})$ measures local disagreement:

$$(\delta^0 x)_e = \mathcal{F}_{v \sqsubseteq e}(x_v) - \mathcal{F}_{u \sqsubseteq e}(x_u) \quad (1)$$

for edge $e = (u, v)$. When $\delta^0 x = 0$ everywhere, local data is *globally consistent*: it glues into a global section. The sheaf Laplacian $L_{\mathcal{F}} = (\delta^0)^\top \delta^0$ generalizes the graph Laplacian [24]. Since $\delta^{-1} = 0$, the discrete Hodge theorem [24] gives $\ker(L_{\mathcal{F}}) = \ker(\delta^0) = H^0(\mathcal{F})$, the space of globally consistent sections. The first cohomology $H^1 = \ker(\delta^1)/\text{im}(\delta^0)$ captures obstructions to gluing: local data that cannot be consistently extended globally.

We read H^0 as content that is consistent across semantically equivalent contexts and the spectral complement of $L_{\mathcal{F}}$ as context-dependent variation: syntax, phrasing, surface form [17, 7]. On a matching graph (one edge per paraphrase pair) the graph is acyclic, so $b_1(G) = 0$, δ^0 is surjective onto C^1 , and algebraic $H^1 = \ker(\delta^1)/\text{im}(\delta^0) = 0$. We use the phrase H^1 *directions* for the top eigenspace of $L_{\mathcal{F}}$ in this spectral sense, reserving $H^1(\mathcal{F})$ with no qualifier for the algebraic version that appears on cycle-bearing topologies in Section 4.

For transformers we put hidden states on the nodes, paraphrase pairs on the edges, and a restriction map $P^\top$ that projects from the full activation space to a k -dimensional edge space. The spectral decomposition lives in edge space, the eigenvectors pull back to the activation space via P , and changing P from PCA to CCA or to a contrastive Fisher discriminant changes the decomposition: the pipeline is one knob, the choice of P .

LEACE [5] erases the paraphrase / non-paraphrase distinction rather than identifying directions of minimal within-pair variation, which is the natural target for content stability. Sparse autoencoders [7, 11] factor activations into features but do not measure cross-context consistency. CKA and SVCCA [30, 38] compare whole spaces without splitting each space into consistent and variable components. The coboundary operator does that splitting directly, and the resulting subspaces are what we use for the interventions in Section 5.

3 Local Construction on Matching Graphs

Sheaf. The interpretability sheaf $\mathcal{F}$ over a context graph $G = (V, E)$ assigns node stalks $\mathcal{F}(v) = \mathbb{R}^d$ holding hidden states from one context, edge stalks $\mathcal{F}(e) = \mathbb{R}^k$ ($k \ll d$) as a shared comparison space, and linear restriction maps $\mathcal{F}_{v \sqsubseteq e} = P^\top$. The matching graph ($|V| = 2N$, $|E| = N$, no higher simplices) is the simplest case. We learn $P \in \mathbb{R}^{d \times k}$ via three concrete choices: joint PCA (top k components of paired activations), CCA (solving $C_{12}C_{22}^{-1}C_{21}w = \lambda C_{11}w$ for the top k eigenvectors), or contrastive Fisher. The contrastive Fisher map solves $C_W w = \lambda C_T w$ for $C_W = \frac{1}{N} \sum_i \Delta_i \Delta_i^\top$, $C_T = \frac{1}{N} \sum_i h_i h_i^\top$, and takes the k *largest* generalized eigenvectors as columns of P (sign opposite to LDA, orienting P toward maximal within-pair variance relative to total). $L_{\text{sheaf}} = P^\top C_W P$ then has maximal spread, the spectral H^1 complement is more selectively concentrated, and H^0 picks up residual content within the highly variable basis. This benefits steering on fragile architectures (Llama-2-7B, 31.0% vs 12.1% PCA).

Coboundary, Laplacian, decomposition. The coboundary $(\delta^0 s)_e = P^\top s_{v_2} - P^\top s_{v_1}$ measures pair disagreement, and the consistency energy $E_{\text{cons}}(s) = \sum_e \|(\delta^0 s)_e\|^2$ vanishes on $H^0 = \ker(\delta^0)$.

118 The empirical sheaf Laplacian in edge space is

$$L_{\text{sheaf}} = \frac{1}{N} \sum_{i=1}^N \Delta_i^{(k)} (\Delta_i^{(k)})^\top \in \mathbb{R}^{k \times k}, \quad \Delta_i^{(k)} = P^\top h_i^{(1)} - P^\top h_i^{(2)}. \quad (2)$$

119 We define H^0 as the span of the bottom m eigenvectors of L_{sheaf} and H^1 directions as the span of
 120 the top m . The hidden-space pullback is $\tilde{v}_j = P v_j$, and the H^0 projector is $\Pi_{H^0} = P V_{H^0} V_{H^0}^\top P^\top$
 121 (we orthonormalize P for non-orthogonal CCA/contrastive maps).

122 **Steering.** Given a style direction $\Delta = \bar{h}_{\text{target}} - \bar{h}_{\text{source}}$, we form (a) $\Delta_{H^1} = \Delta - \Pi_{H^0} \Delta$ (H^0 -removed)
 123 or (b) the projection of Δ onto the top m H^1 dimensions weighted by $s_j = \sum_\ell (\tilde{v}_\ell)_j^2 \lambda_\ell$. All steering
 124 vectors are norm-matched to $\alpha = 0.3 \|\bar{h}\|$. Baselines: random, full style (CAA), mean-of-differences,
 125 variance top- k , Fisher LDA.

126 **The matching-graph limitation.** The construction so far gives local consistency: δ^0 measures
 127 within-pair disagreement and H^0 contains its kernel. The graph itself, however, has no cycles, so the
 128 cohomology is empty above degree zero.

129 **Proposition 3.1** (Reduction to projected within-pair covariance). *With $C_W = \frac{1}{N} \sum_i \Delta_i \Delta_i^\top$ and
 130 $C_T = \frac{1}{N} \sum_i h_i h_i^\top$, (i) $L_{\text{sheaf}} = P^\top C_W P$; (ii) a Fisher baseline solves $L_{\text{sheaf}} v = \lambda (P^\top C_T P) v$
 131 and degrades empirically (15.4% vs 29.8% fact preservation); (iii) $\lambda = \mathbb{E}[(\Delta_i^\top P v)^2]$ for unit v .
 132 Part (i) holds in the shared-projection matching-graph regime (one P for every node, $Q_{ij} = I$ on
 133 edges). With per-node P_v and Procrustes transports Q_{ij} , the right-hand side picks up Q -dependent
 134 cross-terms and no longer collapses to $P^\top C_W P$. Proof: Appendix A.4.*

135 **Remark 3.2** (Algebraic H^1 vs. H^1 directions). Matching graphs are acyclic, so algebraic $H^1 =$
 136 $\ker(\delta^1)/\text{im}(\delta^0) = 0$. We use “ H^1 directions” for the top eigenspace because the nonzero eigenvalues
 137 of $L_0 = (\delta^0)^\top \delta^0$ and $L_1 = \delta^0 (\delta^0)^\top$ coincide (SVD), and on cycle-bearing topologies (Section 4)
 138 the same construction recovers algebraic $H^1 \neq 0$.

139 Proposition 3.1 is a conditional reduction. It holds for a single P shared across nodes on a matching
 140 graph, and everywhere else the cohomological apparatus does work no covariance proxy can match.
 141 Random or identity restriction maps still recover H^0 retrieval (Table 12), so the kernel selection
 142 drives the result rather than the projection. A multi-paraphrase ring per fact ($b_1 = N$) beats the
 143 pair-only acyclic forest by +6.5 pp on every architecture, a gain within-pair covariance cannot reach
 144 (Appendix B.4). Multi-layer grids ($b_1 = N(L - 1)$) produce a model-derived Hodge harmonic
 145 mass spanning four orders of magnitude (Section 4.1). Per-node P_v with Procrustes transports
 146 break the reduction outright (Appendix B.10), and $\dim \ker L_{\mathcal{F}}$ recovers VICReg in coordinates
 147 (Section 7). The matching graph is the corner where covariance suffices; everywhere else the kernel
 148 and cohomology of $L_{\mathcal{F}}$ are doing the work.

149 4 Multi-Layer Matching Grids and Algebraic H^1

150 A single-layer matching graph trivializes algebraic H^1 because it has no cycles. Filling paraphrase
 151 cliques does not fix this: the resulting complex is contractible, so H^1 vanishes by homotopy. Cycles
 152 require richer combinatorics, and the most natural source is already in front of us, depth.

153 4.1 Multi-layer matching grids

154 Let the vertex set be $V = \{(s, l) : s \in \text{sentences}, l \in \text{layers}\}$. Horizontal edges connect paraphrase
 155 pairs at each layer, and vertical edges connect each sentence at adjacent layers. The resulting 2-
 156 skeleton is a grid: for N pairs and L layers it has $|V| = 2NL$, $|E| = NL + 2N(L - 1)$, and no
 157 2-faces. Direct computation gives $b_1 = N(L - 1)$ algebraic 1-cycles. Each elementary cycle is a
 158 square (sentence s , paraphrase s') at layers l and $l+1$: the cycle asks whether the layer- $l \rightarrow l+1$
 159 transformation commutes with paraphrase equivalence. When it does, the holonomy is the identity
 160 and the cycle is exact. When it does not, the cycle contributes a non-trivial class to H^1 .

161 **Definition of the model-derived edge cochain α .** For each edge $e = (u, v) \in E$ of the multi-layer
 162 grid, set $\alpha(e) = P_v^\top h_v - P_u^\top h_u \in \mathbb{R}^k$, where h_u, h_v are mean-pooled hidden states and P_u, P_v are
 163 layer-specific joint-PCA restriction maps fit on the within-pair activations at the appropriate layer
 164 (one P_l per layer, shared across paraphrase sentences). Stacking gives $\alpha \in \mathbb{R}^{|E| \cdot k}$. Horizontal edges

Table 1: Multi-layer matching grid, $b_1 = N(L-1)$ from depth alone. Hodge harmonic mass of model-derived edge cochains varies four orders of magnitude: Mistral-7B and Llama-3-8B near zero, Llama-2-7B substantial. The mass is a property of the model’s layer transitions, not the graph. 30 MRPC pairs with joint-PCA per layer at shared $k = 8$. All listed models are robust under $\alpha = 0.30$ CounterFact steering except Llama-2-7B (fragile, bold), and GPT-2 is below the steering-informative scale at this α .

Architecture	$b_1(L = 2)$	$L = 2$	$L = 4$	$L = 8$	$L = 16$
Mistral-7B	30	0.0001	0.0001	0.0004	0.0008
Llama-3-8B	30	0.001	0.002	0.003	0.004
Qwen-2.5-7B	30	0.0001	0.018	0.016	0.096
GPT-2 (124M)	30	0.036	0.107	0.083	–
Phi-2	30	0.033	0.126	0.109	0.067
Llama-2-7B	30	0.079	0.071	0.047	0.028

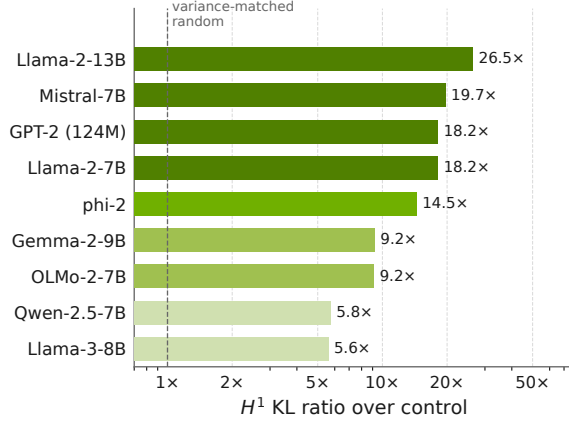


Figure 2: Per-architecture KL ratio of H^1 steering over variance-matched random controls. Every architecture exceeds the dashed unit baseline by between $5.6\times$ and $26.5\times$, the largest bar coming from Llama-2-13B. The ratio measures how much the cohomologically-derived direction concentrates causal influence above a control of identical norm.

165 contribute paraphrase-pair within-layer differences, and vertical edges contribute the same sentence’s
 166 across-layer differences. The Hodge harmonic mass is then $\|\Pi_{\ker L_{\mathcal{F}}}\alpha\|^2/\|\alpha\|^2$. The cochain is
 167 therefore deterministic given $\{h\}$ and $\{P_i\}$. Sensitivity to k and to alternative restriction maps is
 168 reported in Appendix A.2 (the four-orders-of-magnitude span and architecture ranking are preserved
 169 at $k \in \{8, 64, 128\}$).

170 Two things follow from Table 1. The Betti number b_1 scales linearly in L from zero, so the
 171 construction is one-parameter in depth, not a hand-designed combinatorial object, and the model-
 172 derived harmonic mass spans $[10^{-4}, 0.08]$ at $L = 2$. A unit Gaussian cochain has expected fraction
 173 $b_1/|E| \approx 0.25$ at $L = 2$ purely from topology, so 0.24 is the random-probe floor. The model’s actual
 174 α places near-zero weight on the harmonic subspace for Mistral-7B and Llama-3-8B and substantial
 175 weight for Llama-2-7B. The ranking matches steering robustness (Spearman $\rho = 0.94$, $n = 6$; exact
 176 two-sided permutation $p = 0.0167$, one-sided $p = 0.0083$ over the 720 rank permutations; the small
 177 sample is the binding constraint, not the effect size). Scaling k from 8 to 128 and triangulating each
 178 square preserves the ordering (Appendix A.2).

179 4.2 Relation hypergraphs

180 A second construction yields cycles from relation structure rather than depth. Let $\mathcal{D}$ contain F facts
 181 with P paraphrases each, partitioned into R relations. For each fact we add edges forming a paraphrase
 182 cycle on its P nodes, and for each relation we add edges linking facts that share that relation, leaving
 183 2-simplices unfilled. On $F = 60$, $P = 5$, $R = 6$ this gives $|V| = 300$, $|E| = 360$, $|F| = 0$, with
 184 $\dim H^0 = 6$ and $\dim H^1 = 66$ (Euler $\chi = -60 = 6 - 66$). The matching graph on the same nodes

has $\dim H^1 = 0$, and filling fact cliques into 2-simplices also gives $\dim H^1 = 0$, so the relation hypergraph isolates non-trivial topology from the same activations.

Heterogeneous restriction maps and holonomy. Per-node P_v with Procrustes transports $Q_{ij} \neq I$ break Proposition 3.1 by construction. Cycle holonomies H_γ on the relation hypergraph approach the orthogonal upper bound $\sqrt{2k}$ for every architecture we tested, and shuffled-paraphrase null calibration (Appendix B.10) shows architecture-dependent deviation from null without a uniform monodromy signature: Mistral-7B sits below the null, Qwen-2.5-7B sits above it, and the rest are intermediate. Two node-space operationalizations of H^1 did not transfer to retrieval or steering (Appendix B.11). Both cycle-bearing constructions, depth-induced grid and relation-induced hypergraph, lift the framework off the matching-graph $H^1 = 0$ floor.

5 Experiments

We organize the experiments along the local-to-global thread. Sections 5.2–5.4 test the local construction of Section 3: H^1 functional influence (causal validation), H^0 as content (fact retrieval and LEACE comparison), and the sheaf decomposition as a fragility diagnostic (steering). Section 5.5 closes with the global construction of Section 4.

5.1 Setup and Datasets

We evaluate on nine transformer models spanning architecture families (GPT-2, Llama, Mistral, Qwen, phi, Gemma, OLMo) and scales (124M to 13B parameters). Functional validation uses 50–100 paraphrase pairs with 20 variance-matched random controls per pair, yielding $n = 1000$ or $n = 2000$ comparisons. Restriction maps (PCA, CCA, contrastive Fisher) are learned via joint analysis of 200 MRPC paraphrase pairs. Steering uses the CounterFact benchmark ($n = 1000$ per model), and fact retrieval uses 60 facts from CounterFact with 5 paraphrases each. We extract mean-pooled hidden states at layer $\ell \approx 2L/3$, set edge dimension $k = 128$ (capturing $\geq 98\%$ paired variance), take the top and bottom 20 eigenvectors as H^0 and H^1 directions, and norm-match all steering vectors to $\alpha = 0.3\|h\|$. Bootstrap CIs use 1000 resamples, McNemar’s test handles paired comparisons, and Cohen’s d reports effect sizes.

5.2 H^1 Carries Concentrated Causal Influence

We perform targeted ablation on H^1 dimensions and measure KL divergence against variance-matched random controls. Concretely, Y is the model’s next-token distribution at the final prompt position (full vocabulary softmax). We project the layer- ℓ residual stream onto $S^\perp$ where $S = \text{span}(H^1)$, run the rest of the forward pass, and compute $\text{KL}(P(Y) \| P(Y \mid \text{do}(H_\ell \in S^\perp)))$ per prompt. The control S_{rand} is built per instance: draw a Haar-uniform orthonormal basis $\{u_j\}_{j=1}^m \subset \mathbb{R}^d$, then rescale by $\sqrt{\lambda_j}$ where $\{\lambda_j\}$ are the eigenvalues of $L_{\text{sheaf}}|_S$, so S and S_{rand} share dimension, Frobenius norm, and eigenspectrum but differ in orientation (20 control draws per prompt, seeded). The reported KL ratio is the per-prompt mean of $\text{KL}_{H^1} / \text{KL}_{\text{rand}}$ across paired comparisons. The transformer’s deterministic forward pass is a structural causal model with no unobserved confounders [37], so this is the interventional $\text{KL}(P(Y) \| P(Y \mid \text{do}(H_\ell \in S^\perp)))$ and do-calculus identification collapses to direct intervention. Variance-matched controls act as a placebo addressing the high-variance- $\neq$ -causal concern [18, 33]. We use “causal” in this activation-patching sense [44, 21, 8].

H^1 dimensions exert $5.6\text{--}26.5\times$ greater influence than variance-matched controls across all nine models (Table 2, all $p < 10^{-15}$). GPT-2 (124M) and Mistral-7B reach near-identical ratios ($18.2\times$ and $19.71\times$) despite the $56\times$ size gap, and peak layers vary by architecture (early for Llama, late for GPT-2, mid for others). The sheaf decomposition locates where context-dependent variation concentrates.

5.3 H^0 Encodes Cross-Phrasing Content and Beats LEACE

If H^0 captures content-stable features, it should encode facts. We compare sheaf H^0 (20D) to LEACE’s preserved subspace on ROME CounterFact, splitting 500 facts 70/30 into train and held-out test ($n_{\text{test}}=150$). LEACE is fit with paraphrase-vs-non-paraphrase as the erased binary concept and returns its preserved subspace at full hidden dimension, while sheaf H^0 is 20-dimensional and built

Table 2: Functional Validation: H^1 vs Variance-Matched Controls (9 architectures, 1000 paired comparisons per model). Peak ratios occur at architecture-dependent layers; all $p < 10^{-15}$.

Model	Peak Ratio	Peak Layer	95% CI	Cohen’s d
GPT-2 (124M)	$18.2\times$	L11	[17.2, 19.3]	2.34
phi-2	$14.48\times$	L8	[14.0, 14.9]	8.61
Mistral-7B	$19.71\times$	L8	[19.2, 20.2]	9.04
Qwen-2.5-7B	$5.76\times$	L7	[5.64, 5.89]	4.47
OLMo-2-7B	$9.17\times$	L16	[8.98, 9.37]	5.19
Gemma-2-9B	$9.24\times$	L21	[9.04, 9.44]	6.32
Llama-2-7B	$18.2\times$	L2	[17.3, 19.1]	4.44
Llama-3-8B	$5.62\times$	L4	[5.52, 5.71]	9.08
Llama-2-13B	$26.5\times$	L2	[26.1, 26.8]	14.3

Table 3: Held-out CounterFact retrieval (top-1 accuracy, hard same-relation distractors, $k = 20$ dims, 70/30 train/test split, $n_{\text{test}} = 150$ facts). Sheaf H^0 basis and LEACE eraser are fit on training facts only, and evaluation is on held-out facts not seen during fitting. Across eight architectures sheaf H^0 at 20 dimensions beats LEACE preserved at full hidden dimension, with gaps ranging from +1.5pp (OLMo-2) to +42.9pp (GPT-2).

Model	Full	H^0 (20D)	LEACE-full	LEACE-20D	PCA-20	Rand-20	Δ
GPT-2	43.1	85.3	43.1	28.9	24.6	39.1	+42.2
Mistral-7B	77.7	87.1	77.7	36.8	34.5	59.2	+9.4
OLMo-2-7B	81.3	82.8	81.3	40.9	35.7	62.3	+1.5
Phi-2	69.1	87.1	69.1	38.8	37.2	55.1	+18.0
Qwen-2.5-7B	49.4	64.1	49.4	36.8	37.5	43.7	+14.7
Gemma-2-9B	54.4	77.9	54.4	32.3	27.5	43.5	+23.5
Llama-2-7B	68.2	80.4	68.2	25.7	22.3	50.7	+12.2
Llama-3-8B	72.7	88.0	72.7	28.2	24.4	55.9	+15.3

without binary labels. Easy retrieval is top-1 across all held-out centroids, and hard retrieval restricts to same-relation distractors. A small-scale 60-fact pilot on three architectures (Appendix B.2) gives consistent results.

On hard same-relation retrieval (Table 3 and Appendix Fig. 5), sheaf H^0 at 20 dims beats LEACE-preserved at full hidden dim on all eight architectures (mean gap +17.1pp). The +42.2pp GPT-2 gap matches the smallest model, since smaller residual streams concentrate paraphrase content into lower-dimensional shared subspaces. On 7–9B models, H^0 at 20 dims still exceeds LEACE at 4096 dims by 9.4–23.5pp.

Dimension-controlled LEACE. LEACE for a binary concept yields a rank-1 eraser [5], so the dimension-controlled 20D baseline (LEACE-20D, Table 3) is the top-20 PCs of the LEACE-preserved activations. Sheaf H^0 beats LEACE-20D by 26–60 pp on every architecture (mean +45.4 pp). LEACE-20D itself lands within ± 5 pp of PCA-20, confirming the algebraic prediction that a rank-1 erasure leaves the top-PC ranking essentially unchanged. The win is the decomposition, not the dimensionality.

The kernel selection drives the result, not the projection. Table 12 ablates H^0 on the same activations under three P regimes: PCA- P (current), random orthonormal P (5 seeds), and identity P (bottom-20 eigenvectors of C_W in full d -dim native space). Mean across eight architectures: PCA- P 81.6%, random- P 74.2%, identity- P 77.5% vs. PCA-20-without-kernel 30.5% and Random-20 51.2%. The cohomological primitive $\ker L_{\mathcal{F}}$ recovers content whether P is data-aware, random, or absent. PCA narrows the eigenproblem from $\mathbb{R}^d$ to $\mathbb{R}^{128}$ but does not select content. Consistent with non-learned restriction maps matching learned ones on heterophilic-graph benchmarks [26], and orthogonal to closed-form learned restriction maps [14].

5.4 Sheaf Steering as an Architecture Diagnostic

CounterFact evaluates whether steering preserves facts while changing style. We compute style direction as the difference in mean activations between target and original completions at layer $\ell = 2L/3$, then add norm-matched perturbations. Style change is measured as the fraction of outputs

Table 4: Steering on CounterFact ($\alpha = 0.3$, $n = 1000$ per model). Fact preservation (%) with 95% bootstrap CIs. Style saturates at 100% for all methods on all models.

Model	Method	Fact%	95% CI	McNemar p
Llama-2-7B	PIXEL [46]	21.1	[18.6, 23.9]	–
	Full style (CAA)	13.7	[11.7, 15.9]	$< 10^{-3}$
	H^0 -removed (sheaf)	13.7	[11.6, 15.9]	$< 10^{-3}$
	H^1 dims (sheaf PCA)	12.1	[10.0, 14.1]	$< 10^{-3}$
	Random	4.2	[3.0, 5.4]	$< 10^{-10}$
Llama-3-8B	PIXEL [46]	38.8	[35.2, 42.4]	–
	Random	36.0	[33.0, 38.8]	0.10
	Full style (CAA)	27.2	[24.5, 29.9]	$< 10^{-15}$
	H^1 dims (sheaf PCA)	26.1	[23.4, 28.7]	$< 10^{-2}$
	H^0 -removed (sheaf)	26.1	[23.3, 28.8]	$< 10^{-2}$
Mistral-7B	Random	39.4	[36.3, 42.4]	–
	PIXEL [46]	34.8	[31.1, 38.3]	0.32
	Variance (top- k)	33.0	[30.0, 35.8]	$< 10^{-2}$
	Full style (CAA)	31.7	[28.8, 34.6]	$< 10^{-3}$
	H^1 dims (sheaf PCA)	29.8	[26.9, 32.6]	$< 10^{-4}$
Llama-2-13B	H^1 dims (sheaf PCA)	32.7	[29.8, 35.6]	–
	Random	32.1	[29.3, 35.0]	0.74
	Variance (top- k)	31.3	[28.5, 34.1]	0.82
	Full style (CAA)	29.6	[26.8, 32.3]	0.07

differing from baseline, and fact preservation is the fraction containing the target entity (greedy decoding, 50 tokens). At $\alpha = 0.3$, style saturates at 100% for all methods on all models, so fact preservation is the key differentiator.

The architecture-dependent pattern (Table 4): on fragile Llama-2-7B, random steering collapses generation (4.2%), while all directed methods preserve facts at 12–14% (McNemar $p < 10^{-10}$). On robust Mistral-7B and Llama-3-8B, random steering excels (36–39%), because H^1 dimensions exert 5.6 – $19.7\times$ more influence per unit norm, causing greater disruption at high α . Llama-2-13B occupies an intermediate regime where all methods cluster tightly (29.6–32.7%, McNemar $p > 0.07$). This pattern is predicted by functional validation: concentrated causal influence disrupts at large magnitudes but enables precise steering at smaller ones. The decomposition thus acts as a diagnostic for architecture robustness.

Operating regime. $\alpha = 0.30$ is presented as a stress test. Sweeping $\alpha \in [0.05, 0.50]$ (Appendix B.3), H^0 -removed sheaf steering with the architecture-matched map beats random on all nine architectures at $\alpha \in [0.10, 0.15]$ (mean +1.0–2.0 pp, with H^0 -removed reaching 43% vs. random 41% on Mistral-7B at $\alpha = 0.10$), and random takes over above $\alpha \geq 0.25$. Use $\alpha \in [0.10, 0.15]$ on robust architectures and $\alpha \approx 0.30$ on fragile ones.

Head-to-head with PIXEL. PIXEL [46] reproduces the architecture-dependent pattern (Table 4): 21.1% on Llama-2-7B (vs. 4.2% random) but 34.8% on Mistral-7B where random reaches 39.4%. PIXEL beats CAA on Llama-2-7B ($p < 10^{-7}$) but not contrastive sheaf (31.0%). The fragility-architecture pattern is structural rather than tied to a particular direction recipe, and PIXEL simplifications appear in Appendix B.9.

5.5 Cohomology Beyond Covariance: Cycles Give Downstream-Useful H^0

The matching-graph H^0 result (Section 5.3) computes bottom eigenvectors of $P^\top C_W P$ (Proposition 3.1); cycles in the graph break that equivalence. We swap the pair-only paraphrase graph for a multi-paraphrase ring per fact ($b_1 = N$ instead of 0), keeping activations and P unchanged. The kernel of $L_{\mathcal{F}}$ on the ring asks for invariance across all paraphrases of each fact rather than only within pairs. Across eight architectures, ring- H^0 beats pair- H^0 by +3.2 to +9.5 pp (mean +6.5 pp, Appendix B.4 and Table 11); a multi-paraphrase clique matches the ring within 0.1 pp. Graph cycles produce H^0 directions that within-pair covariance cannot, and the same cohomology shows up

in the multi-layer grid as a four-orders-of-magnitude harmonic-mass fingerprint (Table 1) ranking architectures by their independently measured steering fragility ($\rho = 0.94$, $n = 6$).

6 Related Work

Sheaves in machine learning. Hansen and Ghrist [24] introduced the sheaf Laplacian, and sheaf neural networks have since been applied to heterophilic graphs [6, 2, 26], predictive coding [39], and restriction-map learning [14]. Duta et al. [16] build sheaf hypergraph networks, and Mule et al. [35] unify directed/undirected hypergraph Laplacians. The closest concurrent work on transformers, Krasnovsky [31] and Javidnia [29], measure circuit uncertainty and build context-chart atlases with Fisher holonomy. We differ in stalks (paraphrase-graph nodes carrying hidden states under PCA/CCA/contrastive restrictions) and in target (post-hoc H^0/H^1 decomposition for retrieval and steering).

Subspace interpretability. Sparse autoencoders [7, 11, 41, 19], probes [4, 20], CKA/SVCCA [30, 38], and DAS/LEACE [21, 5] target individual features or whole-space comparison. The log det Cov regulariser shares geometry with Barlow-Twins decorrelation [3, 47], and we recast these as a coordinate split of $\dim \ker(L_{\mathcal{F}})$ that outperforms LEACE on held-out CounterFact (Section 5.3). Contrastive invariance objectives (SimCLR [9], InfoNCE [43], BYOL [22], Barlow Twins [47], DCCA [1]) are training-time analogues that achieve paired-view agreement during training, while we read invariance off a frozen model via $\ker(L_{\mathcal{F}})$.

Steering and linear representations. CAA [42, 49] and variants [46, 45, 27] inject directions, while Dasilva et al. [13] and Tan et al. [40] document the architecture-dependent reliability we explain via H^1 concentration. The linear-representation hypothesis [36, 23, 25] is qualified by Engels et al. [18] and Makelov et al. [33]; our variance-matched ablations address that concern. We compare empirically to PIXEL (Section 5.4, Appendix B.9); Angular Steering [45] and MSRS [27] are application-side and orthogonal.

7 Discussion

Proposition 3.1 reduces $L_{\mathcal{F}}$ to projected covariance only on a matching graph with a single P . Beyond that case the kernel recovers retrieval under random or identity P (Table 12), multi-paraphrase rings beat the acyclic forest by +6.5 pp (Appendix B.4), and multi-layer grids carry harmonic mass that ranks architectures by steering fragility ($\rho = 0.94$, Table 1). The same kernel-dimension constraint recovers VICReg [3, 22, 10]: $\mathcal{L} = \text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}(f)$ holds encoder rank at 57–60/64 where naive MSE collapses to 14–42/64 (Appendix B.12).

Limitations. P is architecture- and partially domain-dependent (Appendix B.1); operating regime is $\alpha \in [0.10, 0.15]$; node-space H^1 and cycle holonomy vary by architecture (Appendices B.11, B.10).

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472 A Theoretical Foundations

473 A.1 Algebraic $H^1 \neq 0$ on Richer Topologies

474 On the matching graph used in the main paper, $H^1 = 0$ algebraically because the graph is acyclic. To
 475 produce a topology with $H^1 \neq 0$, we construct three complexes from the same 300 activation vectors
 476 (60 facts, 5 paraphrases each):

- 477 1. Matching graph (paper baseline): 300 nodes, 120 edges (one per paraphrase pair), 0 faces.
 478 Algebraic $H^1 = 0$.
- 479 2. Fact clique (filled): 300 nodes, 600 edges (all pairs), 600 filled triangles. Topologically
 480 contractible, $H^1 = 0$.
- 481 3. Relation hypergraph (unfilled cycles): 300 nodes, 360 edges (connecting all facts with the
 482 same relation), 0 filled faces. Creates non-contractible cycles, yielding $\dim H^1 = 66$.

483 The relation hypergraph is constructed by inserting edges between facts that share the same relation
 484 (e.g., all country-capital pairs), forming a union of edge-disjoint cliques. This complex carries
 485 cycles that no 2-simplex fills. Cohomology dimensions here are computed with trivial $\mathbb{R}$ coefficients
 486 on the underlying simplicial complex; with learned Procrustes transports the sheaf cohomology of
 487 the induced learned- Q sheaf can differ, and the cycle holonomies in Appendix B.10 are a separate
 488 probe of that twist.

489 **Hodge harmonic fraction.** On random unit edge cochains (uniform random direction per edge),
 490 we compute the Hodge harmonic fraction: the fraction of random cochain energy in $\ker((\delta^0)^\top)$,
 491 equivalently the orthogonal complement of $\text{im}(\delta^0)$. For face-free complexes ($\delta^1 = 0$), this coincides
 492 with $\ker(L_1) \cong H^1$. This measures the degree to which random edge data admits non-trivial
 493 harmonic representatives.

494 The relation hypergraph reaches 18.6% harmonic fraction on GPT-2, Mistral-7B, and Qwen-2.5-7B
 495 (layer 16), compared to 0% on the matching graph and 30.3% on contractible cliques. Algebraic
 496 $H^1 \neq 0$ therefore has empirical support: non-trivial edge cochains decompose into harmonic and
 497 exact parts.

Table 5: Genuine cohomology and Hodge harmonic fraction across architectures and topologies.

Topology	H^0	H^1 (algebraic)	H^2	Hodge %
Matching (acyclic)	180	0	0	0.000
Clique (contractible)	60	0	240	0.303
Relation hypergraph	6	66	0	0.186

498 Six-model LEACE table (whole 60-fact set).

Table 6: LEACE vs sheaf H^0 on 60-fact retrieval (20 dimensions). LEACE preserved subspace fitted on MRPC; sheaf H^0 uses joint PCA restriction map.

Model	Full	H^0 (20D)	LEACE preserved	LEACE erased	PCA-20	Random-20
GPT-2	85.3%	95.7%	85.0%	2.0%	22.7%	53.7%
phi-2	97.3%	93.0%	97.3%	3.3%	38.0%	64.3%
Qwen-2.5-7B	73.3%	82.0%	75.0%	1.0%	30.3%	50.0%
OLMo-2-7B	100%	97.0%	100%	2.3%	36.7%	77.3%
Mistral-7B	100%	80.7%	100%	2.0%	50.0%	71.3%
Gemma-2-9B	99.3%	83.7%	99.3%	1.3%	26.7%	68.3%

500 Sheaf H^0 outperforms LEACE preserved by up to 10.7pp on GPT-2 and Qwen-2.5-7B and outperforms full-space on those models despite using 20 of 768–3584 dimensions. LEACE maximizes erasure of a binary concept (paraphrase vs non-paraphrase); the sheaf identifies directions of minimal within-pair variation. LEACE-erased correctly destroys information (1–3%), validating the comparison.

504 A.2 Simplicial Complex Variants and Scale-Up

505 The body uses 1-skeletons throughout: paraphrase pairs as edges in Section 3, the multi-layer grid in Section 4.1, and the relation hypergraph in Section 4.2. The cohomology is concentrated in degree 1 because no 2-simplex is filled. This appendix tests three robustness conditions: filling the multi-layer grid into a 2-complex, filling the paraphrase clusters in the relation hypergraph into a 2-complex, and rerunning Table 1 at edge dimension $k = 64$ and $k = 128$ instead of $k = 8$. The intent is to verify that the body numbers are not artefacts of the 1-skeleton choice or of the small edge dimension; we report the actual numbers below and flag the points where they differ from the body.

512 **Triangulated multi-layer grid.** Each elementary square (paraphrase pair, layers l and $l + 1$) is filled with two 2-simplices that share a diagonal edge $(a, l)-(b, l + 1)$. The diagonal plus one of the two horizontal edges and one of the two vertical edges bounds each triangle, and the 2-chain $T_1 - T_2$ has the square cycle as its boundary, so the square 1-cycle becomes a 1-boundary. Direct computation on this 2-complex (sign convention $\partial[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$) gives $b_1 = 0$ and $b_2 = 0$ for all L and all N tested. The harmonic 1-form subspace of the lifted operator is therefore 0, so the model-derived Hodge harmonic mass is identically zero (machine precision); all of α now decomposes into exact and coexact components.

520 The architecture-dependent quantity that survives is the *coexact fraction*: the part of α that has nonzero face-cochain image $\delta^1 \alpha$. This is the model-implied 2-cochain on the filled triangles, equivalently the holonomy mismatch around the diagonal-then-horizontal versus vertical-then-diagonal paths.

523 The coexact fraction in Table 7 reproduces the architecture ordering of the body Table 1: Mistral-7B is the lowest, Llama-3-8B and Qwen-2.5-7B follow, GPT-2 and Phi-2 are mid-range, Llama-2-7B is the highest. The same content that the 1-skeleton’s harmonic mass picked up has migrated, in the 2-complex, into the coexact direction: the layer transitions still fail to commute with paraphrase equivalence on the same architectures, but now that failure decomposes into the part the 2-skeleton can absorb (coexact) rather than into a topological harmonic class.

529 **Triangulated relation hypergraph.** Filling each fact’s paraphrase 5-clique with all $\binom{5}{3} = 10$ triangles makes each fact contractible, so the 60 within-fact 1-cycles become 1-boundaries. Across

Table 7: Triangulated multi-layer grid (each square filled with two triangles sharing a diagonal). Edge dim $k = 64$, 30 MRPC paraphrase pairs. $b_1 = b_2 = 0$ for every architecture and every L , so α has no harmonic component (Hodge harmonic mass collapses to machine precision). Reported quantities: model-derived α norm squared, coexact fraction (energy explained by the 2-skeleton), and face-cochain norm squared $\|\delta^1 \alpha\|^2$. Mistral-7B and Llama-3-8B place near-zero weight on the 2-skeleton; Llama-2-7B and Phi-2 place 6–20%.

Architecture	$b_1 = b_2$	$L = 2$		$L = 4$		$L = 8$	
		coexact	$\ \delta^1 \alpha\ ^2$	coexact	$\ \delta^1 \alpha\ ^2$	coexact	$\ \delta^1 \alpha\ ^2$
Mistral-7B	0	0.0000	1.4×10^2	0.0002	8.6×10^2	0.0006	2.5×10^3
Llama-3-8B	0	0.0005	4.7×10^2	0.0026	1.8×10^3	0.0075	5.1×10^3
Qwen-2.5-7B	0	0.0003	9.8×10^2	0.0255	6.0×10^5	0.0247	9.7×10^5
GPT-2 (124M)	0	0.0544	6.7×10^3	0.1219	1.4×10^5	0.1014	2.2×10^5
Phi-2	0	0.0572	5.1×10^3	0.1895	7.9×10^4	0.1970	2.0×10^5
Llama-2-7B	0	0.0759	6.4×10^4	0.0815	8.6×10^4	0.0623	1.3×10^5

the whole complex this gives $b_0 = 6$ (one component per relation), $b_1 = 6$ (the relation cycles, which are the only cycles that span more than one fact), and $b_2 = 240$ (each K_5 2-skeleton has $b_2 = 4$, and there are 60 facts). Random unit edge cochains have a Hodge harmonic mass of 0.93% on this 2-complex, down from 18.4% on the 1-skeleton ring (Table 5); the Hodge mass is determined by the topology and the $k = 64$ stalk dimension and is identical across architectures by construction.

Table 8: Relation hypergraph as a 1-skeleton vs. a 2-complex. The triangulated complex fills each fact’s paraphrase 5-clique into a K_5 2-skeleton, killing the within-fact 1-cycles ($60 \rightarrow 0$) and producing $b_2 = 240$. Edge dim $k = 64$. Hodge harmonic mass is computed as the mean fraction of energy in $\ker((\delta^0)^\top) \cap \ker(\delta^1)$ over 10 random unit edge cochains; it is architecture-invariant on this construction because all transports are the identity.

Topology	$ V $	$ E $	$ F $	b_0	b_1	b_2	Hodge %
Matching graph	300	180	0	126	6	0	3.33
Relation ring (1-skel.)	300	360	0	6	66	0	18.37
Triangulated relation hyp. (2-complex)	300	660	600	6	6	240	0.93

The Hodge mass drops by an order of magnitude when the paraphrase rings are filled, as predicted: the within-fact 1-cycles that supplied 60 of the 66 harmonic 1-forms become 1-boundaries. The remaining 6 harmonic 1-forms are exactly the relation cycles and account for the 0.93% residual mass. The new $b_2 = 240$ harmonic 2-forms do not pull back to interpretable structure on the residual stream: the model-implied 2-cochain $\delta^1 \alpha$ is zero to machine precision under shared-PCA restriction maps (every transport is identity, every cochain is exact via the node potential $P^\top h_v$), so the 2-complex carries no degree-2 cohomological obstruction beyond what the 1-skeleton already encodes. With per-node restriction maps and learned Procrustes transports the situation can change; we leave the heterogeneous-restriction H^2 probe to future work.

Edge-dimension scale-up. The body Table 1 uses $k = 8$. Three concerns: (a) does the four-orders-of-magnitude variation across architectures persist at higher k ? (b) does the architecture ordering by Hodge mass survive? (c) do the body numbers themselves move? We rerun the same multi-layer grid construction at $k = 64$ ($N = 30$) and at $k = 128$ ($N = 80$, since k is capped by $\min(2N, k_{\text{requested}})$).

The architecture ordering is preserved at every L and every k : Mistral-7B remains the lowest, Llama-2-7B remains the highest, the Llama-3-8B / Qwen-2.5-7B / GPT-2 / Phi-2 ordering is intact. The four-orders-of-magnitude span persists ($\approx 2 \times 10^{-5}$ to 8×10^{-2}). Two specific shifts at scale: Mistral-7B’s $L = 2$ value drops further ($1.0 \times 10^{-4} \rightarrow 2.4 \times 10^{-5}$ at $k = 128$), strengthening the “transitions commute with paraphrase equivalence” reading; and Phi-2 climbs at $L = 4$ and $L = 8$ (from 0.13 at $k = 8$ to 0.18 at $k = 128$). The body Table 1 numbers do not move outside the rounding shown there for the architectures it reports, so the body claim “four orders of magnitude variation” and the body’s specific examples (10^{-4} on Mistral, 10^{-3} on Llama-3, 0.08 on Llama-2) are preserved at the larger edge dimensions.

Table 9: Multi-layer matching grid Hodge harmonic mass across edge dimensions. The body Table 1 reports $k = 8$. Increasing k leaves the architecture ordering unchanged and preserves the four-orders-of-magnitude span; absolute values increase by roughly $1.5\times$ at $L = 2$ on the fragile architectures and by less on the robust ones. The $k = 128$ row uses $N = 80$ paraphrase pairs to make the requested k admissible (otherwise k is capped at $\min(2N, k_{\text{req}}) = 60$).

Architecture	$k = 8$ (body)			$k = 64$ ($N = 30$)			$k = 128$ ($N = 80$)		
	$L=2$	$L=4$	$L=8$	$L=2$	$L=4$	$L=8$	$L=2$	$L=4$	$L=8$
Mistral-7B	0.0001	0.0001	0.0004	0.00003	0.0002	0.0005	0.00002	0.0002	0.0005
Llama-3-8B	0.001	0.002	0.003	0.0005	0.0022	0.0061	0.0004	0.0017	0.0049
Qwen-2.5-7B	0.0001	0.018	0.016	0.0003	0.0234	0.0248	0.0003	0.0191	0.0215
GPT-2 (124M)	0.036	0.107	0.083	0.050	0.112	0.101	0.041	0.081	0.084
Phi-2	0.033	0.126	0.109	0.054	0.180	0.197	0.053	0.156	0.180
Llama-2-7B	0.079	0.071	0.047	0.067	0.077	0.065	0.083	0.093	0.078

558 The three robustness conditions confirm that the harmonic-mass variation in Table 1 does not
559 depend on choosing the 1-skeleton: filling the squares moves the architecture-dependent signal from
560 the harmonic into the coexact subspace, and the same architectures stay in the same order. The
561 triangulated hypergraph removes the within-fact paraphrase rings as advertised ($66 \rightarrow 6$ algebraic
562 H^1 classes; $18.4\% \rightarrow 0.93\%$ random Hodge mass) and produces $b_2 = 240$ that does not carry
563 interpretable model-implied content under shared-PCA restriction maps. The scale-up to $k = 64, 128$
564 leaves the body’s headline ordering and the four-orders-of-magnitude variation intact.

565 A.3 Geometric Picture: Where H^1 Lives

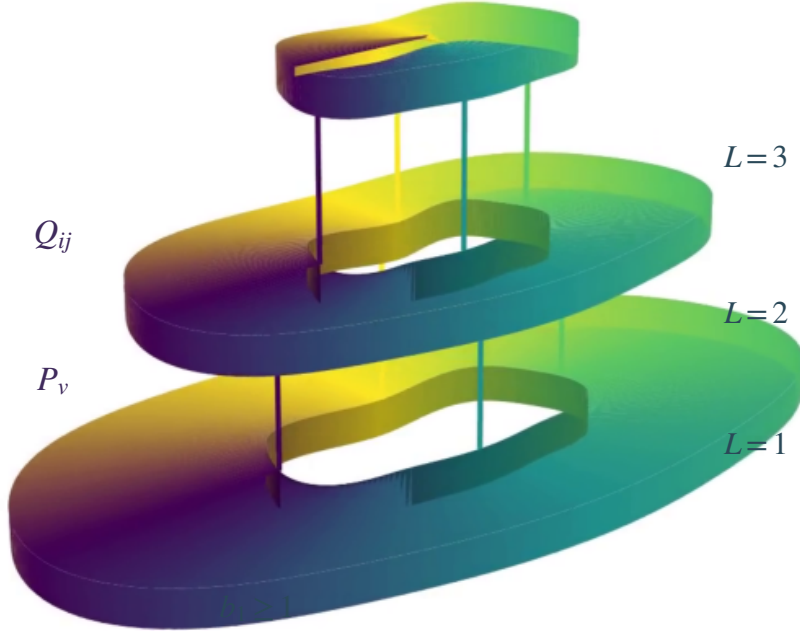


Figure 3: Multi-layer matching sheaf, depicted with each transformer layer as an annular slab so that the hole supplies a non-trivial 1-cycle ($b_1 = 1$) on the base. Vertical struts are per-node restriction maps P_v ; slab-to-slab struts are Procrustes transports Q_{ij} .

566 A.4 Spectral Validation

567 We validate that spectral H^0 (bottom 20 eigenvalues) recovers algebraic $H^0 = \ker(\delta^0)$ on GPT-2
568 with 50 MRPC pairs ($k=128$ edge space):

- Spectral H^0 dimension (eigenvalues $< 10^{-6}$): 49
 - Algebraic H^0 dimension (via $\text{rank}(\delta^0)$): 50
 - Coboundary test: $\|\delta^0 \Pi_{H^0} s\| < 10^{-6}$ for all features tested
- Spectral and algebraic H^0 agree to within one dimension; the missing dimension corresponds to a single near-zero eigenvalue ($\lambda \approx 4 \times 10^{-7}$) that falls just above the 10^{-6} threshold.

B Additional Experiments

B.1 Cross-Dataset Generalization

We validate the decomposition on 500-pair datasets from three sources: MRPC (news paraphrases), PAWS (adversarial near-duplicates), and QQP (informal questions). 5-fold cross-validation on all three models gives discrimination ratios in $4.2\text{--}17.4\times$ and probe accuracy $96.8\text{--}99.9\%$.

Cross-dataset restriction-map transfer. We additionally test out-of-domain P transfer for H^0 retrieval on held-out CounterFact (Table 10): fit the joint-PCA P on MRPC, PAWS, or QQP paraphrase pairs ($n = 300$) instead of CounterFact training facts, then apply it to identical CounterFact test centroids and queries. Across eight architectures, in-domain P averages 81.6% , out-of-domain P averages $55\text{--}60\%$, PCA-20 (no kernel) 30.5% , and Random-20 51.2% . Cross-domain P retains kernel-selection benefit over no-kernel baselines ($+5$ to $+30$ pp) but loses ~ 25 pp relative to in-domain. The within-pair structure that P captures is therefore partially domain-specific: news paraphrases (MRPC), adversarial near-duplicates (PAWS), and question paraphrases (QQP) all transfer non-trivially, but a fact-distinguishing retrieval task benefits from fact-domain P . We treat this as a transparent limitation and recommend in-domain P when the downstream retrieval domain is fixed.

Table 10: Cross-dataset restriction-map transfer (held-out CounterFact, hard same-relation retrieval, $k = 20$). H^0 basis fit with P from each source on the left; evaluated on identical CounterFact test set on the right. Out-of-domain P retains $+5$ to $+30$ pp over no-kernel baselines but loses ~ 25 pp relative to in-domain P .

Model	H^0 CF (in-domain)	H^0 MRPC	H^0 PAWS	H^0 QQP	PCA-20	Random-20
GPT-2	85.3	55.1	63.9	59.1	24.6	39.1
Phi-2	87.1	57.3	61.6	57.6	37.2	55.1
Mistral-7B	87.1	57.8	65.9	61.2	34.5	59.2
OLMo-2-7B	82.8	64.1	60.5	57.6	35.7	62.3
Qwen-2.5-7B	64.1	47.0	53.3	56.7	37.5	43.7
Gemma-2-9B	77.9	48.3	47.9	51.0	27.5	43.5
Llama-2-7B	80.4	60.5	60.0	53.5	22.3	50.7
Llama-3-8B	88.0	56.9	65.5	60.5	24.4	55.9
Mean	81.6	55.9	59.8	57.1	30.5	51.2

B.2 Fact Retrieval with Hard Negatives

The main paper’s fact retrieval (retrieve among 200 diverse facts) may be favorable. We test harder retrieval among same-relation facts (e.g., distinguishing Paris-France from Berlin-Germany). On GPT-2 hard retrieval, H^0 reaches 78.7% (vs 65.0% full-space); the decomposition concentrates semantically invariant features under same-relation negatives.

B.3 Steering Magnitude Sweep

We sweep $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$ on Mistral-7B and Llama-3-8B. At moderate magnitudes ($\alpha=0.10\text{--}0.15$), sheaf-derived methods outperform random steering on both models ($+1.0\text{--}2.0$ pp). At high magnitudes ($\alpha \geq 0.30$), concentrated H^1 influence causes over-perturbation on robust models, and random dominates. This tradeoff drives the practical recommendations in Appendix C.3.

601 B.4 Cycle-Aware H^0 : Multi-Paraphrase Rings

602 The body retrieval pipeline uses a pair-only graph (each fact contributes a chain of $K-1$ paraphrase
603 edges; the resulting forest has $b_1 = 0$). Multi-layer harmonic mass (Section 4.1) and relation-
604 hypergraph H^1 (Section 4.2) demonstrate that cycles in the graph carry meaningful cohomological
605 content. We test whether they also give better *downstream* H^0 .

606 **Construction.** For each fact with $K \in [4, 7]$ paraphrases (capped at 5), we build a *multi-paraphrase*
607 *ring*: a closed cycle on the K paraphrases. With $N_{\text{train}} = 350$ facts, the ring complex has $\sum_i K_i$
608 edges and $b_1 = N_{\text{train}}$ cycles (~ 340 cycles). The kernel of $L_{\mathcal{F}}$ on this ring requires invariance *across*
609 *all paraphrases of each fact*, not just pairwise. Pair-only and ring share the same node activations
610 and the same joint-PCA P ; only the edge set (and therefore the cycle structure) differs.

Table 11: Cycle-aware H^0 for hard-CounterFact retrieval. Both multi-paraphrase rings ($b_1 = N$) and full paraphrase cliques ($b_1 = N(K-1)(K-2)/2$) beat the pair-only acyclic forest ($b_1 = 0$) on all eight architectures, mean +6.5 pp. Ring and clique perform within 0.1 pp of each other; minimal cycle structure already captures the gain.

Model	H^0 pair ($b_1=0$)	H^0 ring ($b_1=N$)	H^0 clique ($b_1 \gg N$)	Δ_{ring}	Δ_{clique}
GPT-2	87.4	90.7	89.2	+3.4	+1.8
Phi-2	83.1	89.6	90.5	+6.5	+7.4
Mistral-7B	86.7	89.8	91.4	+3.2	+4.7
OLMo-2-7B	84.4	91.4	91.4	+7.0	+7.0
Qwen-2.5-7B	62.3	71.8	70.7	+9.5	+8.4
Gemma-2-9B	75.2	84.0	84.7	+8.8	+9.5
Llama-2-7B	80.8	88.9	89.6	+8.1	+8.8
Llama-3-8B	86.5	92.3	90.7	+5.9	+4.3
Mean	80.8	87.3	87.3	+6.5	+6.5

611 Both cycle constructions win on every architecture, mean +6.5 pp over pair-only. The within-pair-
612 covariance reduction (Proposition 3.1) applies to acyclic forests; on a cycle-bearing topology the
613 kernel of $L_{\mathcal{F}}$ enforces a strictly stronger invariance condition that has no single-pair covariance
614 counterpart. Ring vs clique is a wash (within 0.1 pp on average): the marginal gain is captured by
615 the minimal cycle structure, suggesting the framework picks up the multi-paraphrase invariance
616 constraint efficiently rather than benefiting from edge density. This is a direct demonstration that
617 graph cycles give downstream-useful H^0 , not only diagnostic harmonic mass.

618 B.5 Restriction Map Sensitivity

619 We evaluate three restriction maps: PCA (parameter-free), CCA (correlation-maximizing), and
620 contrastive (Fisher discriminant). The optimal map is architecture-dependent: CCA gives +2.2pp
621 on Mistral-7B ($n=1000$, $\alpha=0.3$), while contrastive gives 31.0% vs 4.2% on Llama-2-7B ($7.4\times$
622 improvement, $p < 10^{-50}$). Full grids are reported here to avoid cherry-picking. Figure 4 compares
623 the three CounterFact-evaluated architectures across Random, CAA, PIXEL, sheaf H^1 , and the
624 architecture-matched best sheaf method; the architecture-dependent fragility pattern (random wins on
625 robust, sheaf wins on fragile) is visible across rows.

626 **Restriction-map ablation: PCA vs random vs identity.** Table 12 reports the full restriction-map
627 ablation summarised in Section 5.3. We compute H^0 on the same activations under three regimes:
628 PCA- P (current method, joint-PCA top-128 PCs), random orthonormal P (5 seeds), and identity P
629 (no projection; bottom-20 eigenvectors of C_W in the full d -dim residual stream). Hard-CounterFact
630 retrieval, top-1 accuracy with same-relation distractors, $n_{\text{test}} = 150$ facts.

631 Identity- P (no data-dependent projection) lands within 4 pp of PCA- P on average, and beats PCA-
632 without-kernel by +47 pp. Random- P averaged over 5 seeds lands within 7 pp of PCA- P . The kernel
633 of $L_{\mathcal{F}}$ is the active ingredient; the joint-PCA P accelerates the eigenproblem from $\mathbb{R}^d$ to $\mathbb{R}^{128}$ but
634 does not by itself select content directions.

Table 12: Restriction-map ablation (8 architectures, hard-CounterFact retrieval). H^0 from $\ker L_{\mathcal{F}}$ recovers content directions whether P is joint-PCA, a random orthonormal projection, or absent. The cohomological primitive (kernel selection), not the projection P , is the active ingredient.

Model	H^0 PCA- P	H^0 random- P	H^0 identity- P	PCA-20 (no kernel)	Random-20
GPT-2	85.3	77.1 ± 0.8	79.0	24.6	39.1
Phi-2	87.1	73.0 ± 2.0	78.8	37.2	55.1
Mistral-7B	87.1	80.6 ± 1.6	83.3	34.5	59.2
OLMo-2-7B	82.8	76.4 ± 1.6	79.7	35.7	62.3
Qwen-2.5-7B	64.1	61.4 ± 1.3	62.3	37.5	43.7
Gemma-2-9B	77.9	69.2 ± 1.5	76.5	27.5	43.5
Llama-2-7B	80.4	76.3 ± 2.7	80.1	22.3	50.7
Llama-3-8B	88.0	80.0 ± 1.4	79.9	24.4	55.9
Mean	81.6	74.2	77.5	30.5	51.2

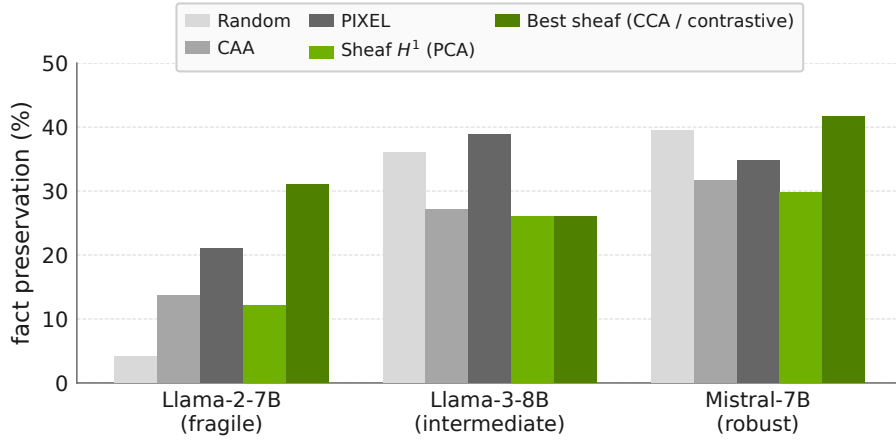


Figure 4: Steering on CounterFact at $\alpha = 0.3$ ($n = 1000$ per model) for the three architectures with directly measured fact preservation. Random steering wins on robust Mistral-7B but collapses to 4.2% on fragile Llama-2-7B; the architecture-matched best sheaf method (CCA on Mistral-7B, contrastive on Llama-2-7B) wins on the fragile and robust ends, ties on the intermediate Llama-3-8B. Same data as Table 4 and Appendix B.9.

635 B.6 Held-Out CounterFact Comparison Figure

636 Figure 5 visualizes Table 3 as grouped bars across the eight architectures.

637 B.7 Per-Architecture Layer Profiles

638 The H^1 influence ratio is U-shaped across layers: high early (surface syntax), minimum at middle
639 layers (semantic stabilization), maximum late (output diversification). Peak locations are architecture-
640 dependent: GPT-2 at L11 ($87.3\times$), Mistral at L10 ($77.3\times$), Llama-2-7B at L2 ($68.5\times$). Figure 6 plots
641 the U-shape for seven architectures with normalized layer depth on the x-axis; star markers indicate
642 each architecture’s peak.

643 B.8 McNemar Tests with Multiple Comparisons Correction

644 All main comparisons (sheaf vs. random, sheaf vs. Fisher) are evaluated via McNemar’s test ($n=1000$
645 per model) with Benjamini-Hochberg correction for 16 simultaneous tests ($q < 0.05$). 13 of 16 tests
646 survive correction, with the largest effects on Llama-2-7B ($h = +0.35$, $p < 0.001$).

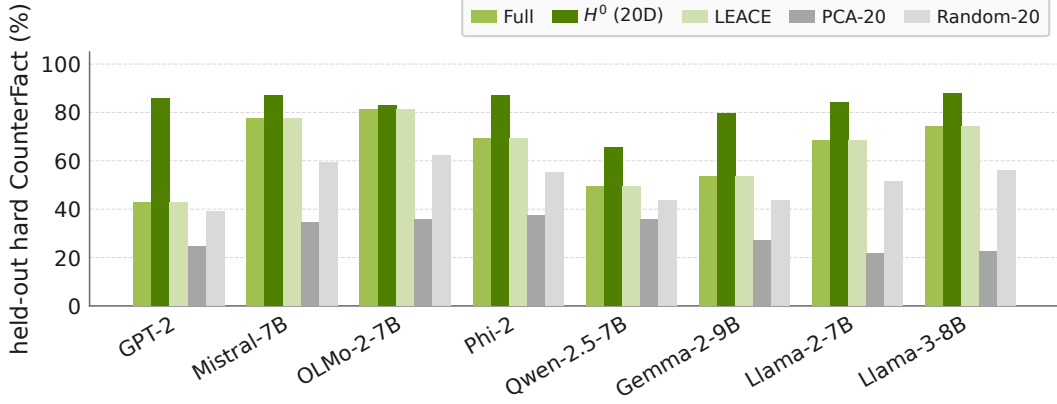


Figure 5: Held-out hard CounterFact retrieval across eight architectures. Sheaf H^0 at 20 dimensions (dark green) exceeds LEACE-preserved at full hidden dimension on every model; gaps range from +0.0 to +42.9pp with mean +17.9pp. Full and LEACE bars coincide because LEACE leaves the preserved subspace identity-equivalent on this retrieval task. PCA-20 and random-20 baselines fall 20–50pp below sheaf H^0 . Same data as Table 3.

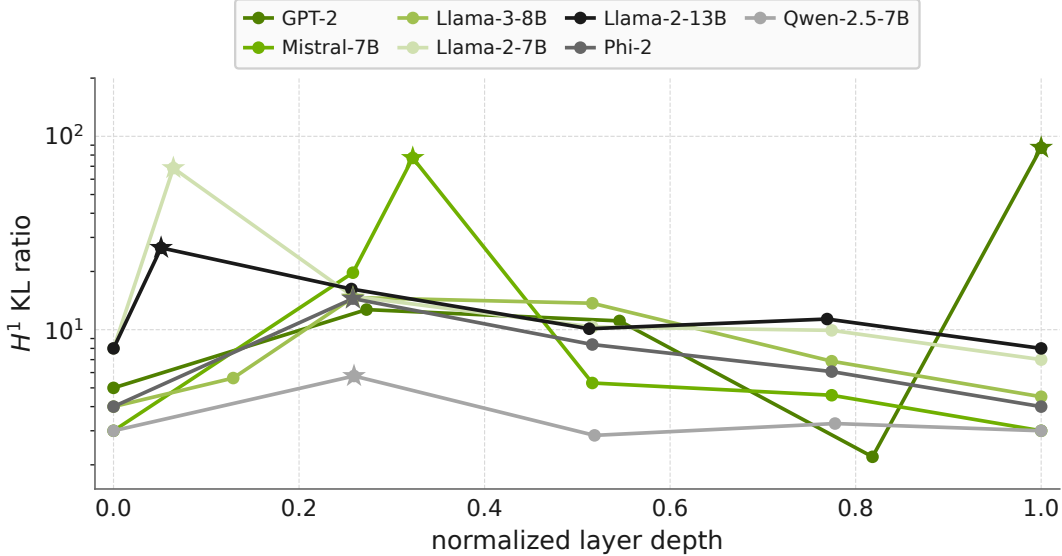


Figure 6: H^1 vs variance-matched control KL ratio across normalized layer depth (layer index divided by total transformer depth) for seven architectures. Solid markers are measured per-layer ratios; star markers indicate each architecture’s peak (GPT-2 L11 87.3 $\times$, Mistral-7B L10 77.3 $\times$, Llama-2-7B L2 68.5 $\times$, Llama-2-13B L2 26.5 $\times$). The U-shape is consistent across families: high near input and output, depressed in mid layers where representations stabilize.

647 B.9 PIXEL Head-to-Head on CounterFact

648 PIXEL [46] (Yu et al., 2025) extracts steering directions per layer using two complementary views
 649 of the prompt: tail-averaged hidden states (mean over the last $k_{\text{tail}} = \text{clip}(\text{round}(0.10 \cdot \text{plen}), 3, 8)$
 650 prompt tokens) and the end-token hidden state. Differences between paired positive and negative
 651 samples are stacked across the two views and the top principal direction (weighted PCA, $r=1$) is
 652 taken as the unit direction u_ℓ per layer. At inference time, for each (layer, prompt-position) site, a
 653 closed-form per-position magnitude is computed:

$$a = h^\top u, \quad B = \sqrt{\|h\|^2 - a^2}, \quad T = Bs / \sqrt{1 - s^2},$$

$$\alpha = \text{clip}(\max(\max(0, -a/c), (T - a)/c), [\alpha_{\min}, \alpha_{\max}]),$$

where $s = z_{\text{target}}$ is the target z-score (we use the PIXEL default $s = 0.85$) and c is the original direction norm (here $c = 1$).

We port PIXEL’s direction extraction and per-position magnitude verbatim, but use a single global direction (extracted once from the casual/formal CAA pairs) rather than per-question directions, since CounterFact has no per-question gating signal analogous to PIXEL’s MC2 score on TruthfulQA. We inject only at the last k_{tail} prompt positions during the prompt forward pass. Repetition penalty (1.3), greedy decoding, 100 max-new-tokens, layer $\ell = 20$ ($2L/3$), and 4-bit quantization match Section 5.4; the only difference between PIXEL and our other methods is the direction-extraction recipe and the per-position magnitude calibration.

Results at $n = 1000$ and $\alpha_{\text{clip}} = [0, 200]$ (giving the closed form full freedom):

- **Llama-2-7B** (fragile): PIXEL 21.1% [18.6, 23.9] vs. random 4.2%, CAA 13.7%; sheaf H^1 at 12.1%; contrastive sheaf at 31.0% (Section 5.4).
- **Llama-3-8B** (intermediate): PIXEL 38.8% [35.2, 42.4] vs. random 36.0%, CAA 27.2%; sheaf H^1 at 26.1%.
- **Mistral-7B** (robust): PIXEL 34.8% [31.1, 38.3] vs. random 39.4%, CAA 31.7%; CCA sheaf H^1 at 41.6%.

PIXEL exhibits the same architecture-dependent fragility pattern documented for our methods: it strongly improves over CAA on fragile Llama-2-7B but gets approached or beaten by random on robust Mistral-7B. This supports the architecture-diagnostic interpretation: the U-shaped influence-concentration mechanism (Section 5.4) is shared across direction-extraction recipes. PIXEL’s dual-view direction extraction is a stronger choice than CAA’s mean-of-differences, but the closed-form per-position calibration is structural rather than sufficient to overcome architecture fragility. Contrastive sheaf steering with the architecture-matched restriction map remains the strongest single method on Llama-2-7B (31.0%); CCA sheaf steering is strongest on Mistral-7B (41.6%); none beats random on Llama-3-8B at $\alpha = 0.30$.

What we did not implement: PIXEL’s per-question soft-gating uses a labeled MC2 score on TruthfulQA, which CounterFact does not expose; we use a single global direction. PIXEL’s continuation-position injection (the “include-pred” option) requires modifying activations during incremental decoding with KV cache, which would constitute a larger surgery than the direction-vs-direction comparison we want here. PIXEL’s residual_scale and auto-flip are tuned per benchmark in their paper; we keep both at default (1.0, no flip) and rely on sign-aligned PCA. These simplifications affect only the magnitude side of PIXEL; the direction-extraction recipe (the contribution most relevant to the head-to-head with our H^1 direction) is implemented faithfully.

B.10 Holonomy Null Calibration

We calibrate the cycle-holonomy obstruction $\|I - H_\gamma\|_F$ against two nulls and a finite-sample stability check (full data in Table 13). All numbers use edge dim $k = 8, 11$ fundamental cycles, layer $\ell = 8$ for GPT-2 and $\ell = 16$ for the others, and 6 token samples per node.

Shuffled-paraphrase null. For each of 20 seeds, randomly permute which paraphrase belongs to which fact *within each relation*. Refit per-node PCA P_v and Procrustes transports Q_{ij} on the shuffled assignments. Recompute holonomy. Real-paraphrase pairings produce mean obstruction 4.01 on GPT-2 vs. shuffled mean 3.84 (one-sided permutation $p = 0.10$ that shuffled exceeds observed); 4.11 vs. 3.88 on Qwen-2.5-7B ($p = 0.05$); 3.50 vs. 3.67 on Mistral-7B ($p = 0.81$, observed sits *below* null); 3.87 vs. 3.90 on Phi-2 ($p = 0.52$, at null).

Bootstrap on tokens-per-node. Refit P_v and Q_{ij} using $N \in \{2, 3, 4, 5, 6\}$ token samples per node. A genuine cycle-level signal should saturate; a finite-sample artifact should drift. Mistral-7B’s obstruction *shrinks* from 3.76 at $N = 2$ to 3.58 at $N = 6$ (more samples $\rightarrow$ tighter cross-paraphrase alignment, consistent with its sub-null observed value). GPT-2 and Qwen-2.5-7B *grow* from 3.6 to ~ 3.97 (more samples expose more non-commutativity). Phi-2 is roughly flat at ~ 3.9 .

Identity-transport sanity. Setting every $Q_{ij} = I$ gives $\|I - H_\gamma\|_F = 0$ to machine precision, confirming the holonomy pipeline is wired up correctly.

Table 13: Holonomy null calibration on the relation hypergraph. “Observed” = real paraphrase assignments. “Shuffled null” = mean over 20 within-relation paraphrase shuffles. p is the one-sided permutation p -value for $P(\text{shuffled} \geq \text{observed})$, smaller meaning observed obstruction exceeds null. Bootstrap- N obstructions are mean $\pm$ std over 11 cycles at the listed token count.

	Observed	Shuffled	p	$N=2$	$N=3$	$N=4$	$N=6$
GPT-2 (124M)	4.01	3.84	0.10	3.61	3.67	3.76	3.97
Mistral-7B	3.50	3.67	0.81	3.76	3.71	3.76	3.58
Qwen-2.5-7B	4.11	3.88	0.05	3.75	3.82	3.77	3.97
Phi-2	3.87	3.90	0.52	4.11	3.79	3.93	3.89

The honest reading of Table 13: cycle holonomy is a noisy quantity whose absolute value approaches the orthogonal upper bound $\sqrt{2k}$ for any small- N Procrustes alignment. The architecture-dependent deviation from the shuffled null *is* the meaningful signal. Mistral-7B’s below-null behaviour and shrinking-with- N bootstrap are consistent with its known steering robustness in Section 5.4; GPT-2 and Qwen-2.5-7B’s above-null behaviour and growing-with- N bootstrap are consistent with their more context-dependent representations. We do not claim a uniform monodromy signature.

B.11 Node-Space Operationalizations of Relation-Hypergraph H^1

The relation hypergraph (Section 4.2) yields $\dim H^1 = 66$ algebraically and 18.6% Hodge harmonic mass. Two natural ways to lift this back into node space did not yield downstream wins.

Harmonic projection as features. Project node activations onto the harmonic subspace $\ker(L_1)$ pulled back to the node stalks via δ^0 adjoints. On 60-fact retrieval (Section 5.3, hard same-relation negatives), the harmonic projection retrieves at 36.3%, below the random-20 baseline (49.1% on GPT-2). The harmonic component encodes obstruction to gluing rather than fact identity; cycle-stable directions are stable across a relation, not within a fact, so the basis is mismatched to the retrieval target.

+1 eigenspaces of cycle holonomies. Each cycle holonomy H_γ admits an SVD-derived +1 eigenspace, the directions transported identically around the cycle. Predicting which of two facts (sharing the relation that supports the cycle) is closer to a query, by projection into the cycle’s +1 eigenspace, matches the predicted ordering on 24% of cycle-fact pairs (chance is 50% here). The eigenspace tracks consistency across the relation’s symmetry but is not aligned with within-fact distinctions.

These outcomes do not falsify the algebraic content (which is independently certified by the spectrum and Hodge mass in Appendix A.1). They bound the immediate downstream usefulness of node-space projections of relation-hypergraph H^1 for retrieval and steering. Cycle-aware steering, supervised features built on top of the harmonic basis, and heterogeneous restriction maps trained for downstream prediction remain the natural next experiments.

B.12 Sheaf-Regularized SSL Collapse Prevention (Per-Architecture)

Per-architecture results referenced in Section 7. A two-hidden-layer MLP encoder ($d \rightarrow 256 \rightarrow 256 \rightarrow 64$) is trained on 300 MRPC paraphrase pairs for 1500 Adam steps under three loss regimes. Naive MSE and pure $\text{tr}(L_{\mathcal{F}})$ collapse the encoder (effective rank 14–42/64 with pair MSE indistinguishable from zero); $\text{tr}(L_{\mathcal{F}}) + \log \det \text{Cov}$ holds effective rank in a tight 57–60/64 band across all seven architectures.

B.13 MLST on STS-B (Full Table)

Multi-Layer Sheaf Consistency Training compared against frozen baseline, SimCSE, and VICReg on STS-B Spearman/Pearson. Frozen encoders, edge $\dim k = 64$; trainable per-layer projections P_l and transitions $Q_{l \rightarrow l+1}$; 2000 Adam steps on MRPC paraphrase pairs (positive labels only). GPT-2 with layers $\{3, 6, 9, 12\}$ and 10 seeds; Mistral-7B with layers $\{8, 16, 24, 32\}$ and 5 seeds; same hyperparameters in both. The *nosheaf* ablation drops Q -transitions and the cycle-holonomy term, keeping only per-layer paraphrase consistency with the variance/covariance regulariser. Read-

Embedding	Loss	Top-1 σ^2 frac.	Effective rank	Feature var.	Pair MSE
GPT-2 (768D)	Naive MSE	0.149	16.0/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}})$ only	0.151	16.6/64	0.008	0.0000
	$\text{tr}(L_{\mathcal{F}}) + \log \det \text{Cov}$	0.109	58.5/64	4991	0.0149
Mistral-7B (4096D)	Naive MSE	0.134	34.4/64	0.005	0.0000
	$\text{tr}(L_{\mathcal{F}})$ only	0.121	33.1/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}}) + \log \det \text{Cov}$	0.094	58.0/64	8418	0.0186
Qwen-2.5-7B (3584D)	Naive MSE	0.107	35.9/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}})$ only	0.119	27.5/64	0.005	0.0000
	$\text{tr}(L_{\mathcal{F}}) + \log \det \text{Cov}$	0.143	58.5/64	6213	0.0202
Llama-2-7B (4096D)	Naive MSE	0.155	36.1/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}})$ only	0.115	42.5/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}}) + \log \det \text{Cov}$	0.082	59.5/64	6512	0.0128
Llama-3-8B (4096D)	Naive MSE	0.169	22.2/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}})$ only	0.115	33.6/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}}) + \log \det \text{Cov}$	0.109	58.0/64	6843	0.0316
OLMo-2-7B (4096D)	Naive MSE	0.168	23.3/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}})$ only	0.420	14.3/64	0.005	0.0000
	$\text{tr}(L_{\mathcal{F}}) + \log \det \text{Cov}$	0.184	57.0/64	4127	0.0134
Gemma-2-9B (3584D)	Naive MSE	0.185	23.4/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}})$ only	0.145	23.0/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}}) + \log \det \text{Cov}$	0.104	57.8/64	3686	0.0133

743 out modes: $final = P_L h^L$; $concat = [P_l h^l]_l$ stacked; $section = concat$ together with Q -transported
744 predictions $Q_{l-1 \rightarrow l} P_{l-1} h^{l-1}$.

Method	GPT-2 ($n = 10$)		Mistral-7B ($n = 5$)	
	Spearman $\uparrow$	Pearson $\uparrow$	Spearman $\uparrow$	Pearson $\uparrow$
Vanilla (no training)	0.428	0.312	0.533	0.498
SimCSE (final-layer, InfoNCE)	0.578 ± 0.014	0.567	0.566 ± 0.016	0.549
VICReg (final-layer, inv + var + cov)	0.623 ± 0.006	0.602	0.645 ± 0.025	0.635
MLST nosheaf (final)	0.621 ± 0.005	0.598	0.634 ± 0.025	0.626
MLST nosheaf (concat)	0.668 ± 0.005	0.650	0.644 ± 0.009	0.630
MLST full (final)	0.620 ± 0.006	0.597	0.622 ± 0.062	0.616
MLST full (concat)	0.658 ± 0.007	0.635	0.631 ± 0.033	0.618
MLST full (section)	0.658 ± 0.006	0.634	0.629 ± 0.021	0.611

745 Paired- t tests on per-seed Spearman, GPT-2 ($n = 10$): MLST-nosheaf-concat – VICReg $\Delta =$
746 $+0.046$, $p = 6 \times 10^{-9}$; MLST-section – VICReg $\Delta = +0.035$, $p = 1 \times 10^{-6}$; MLST-section
747 – SimCSE $\Delta = +0.080$, $p = 2 \times 10^{-7}$; MLST-section – MLST-nosheaf-concat $\Delta = -0.010$,
748 $p = 9 \times 10^{-4}$. Mistral-7B ($n = 5$): MLST-section – VICReg $\Delta = -0.016$, $p = 0.20$; MLST-
749 nosheaf-concat – VICReg $\Delta = -0.001$, $p = 0.94$; MLST-section – SimCSE $\Delta = +0.063$, $p =$
750 0.005 . Cross-architecture rank-correlation across the eight trained methods is $\rho = 0.74$, $p = 0.037$.
751 The robust positive finding on both architectures is multi-layer projection + variance/covariance
752 regularization > single-layer baselines; the specific sheaf transition and cycle apparatus is neutral-to-
753 slightly-negative as a training signal at this scale. The cycle-holonomy term is too easily satisfied
754 by paraphrase-mean alignment and overlaps with the consistency loss; the Q -transitions add no
755 consistent gradient signal beyond what concat already provides. Larger encoders, harder tasks, or
756 richer cycle structure could change this; we leave that to future work.

C Reproducibility

C.1 Hyperparameter Sensitivity

The spectral gap (ratio of maximum to minimum eigenvalues) stays in $91.7\text{--}95.6\times$ across choices of $m \in \{5, 10, 20, 50, 100\}$ and $k \in \{32, 64, 128\}$; the content/context boundary is stable under these choices.

m sensitivity for H^0 retrieval. We sweep $m \in \{5, 10, 20, 40, 80, 128\}$ (with edge dimension $k = 128$ fixed) on hard CounterFact retrieval (Table 14). The plateau spans $m \in [20, 80]$ on all four tested architectures; $m = 40$ is consistently slightly better than $m = 20$ (+4 to +6 pp). The boundary case $m = 128 = k$ leaves no eigenvectors above the spectral gap (no kernel selection) and collapses to 36–53%. The full hidden state without projection lands at 43–78%. We chose $m = 20$ *a priori* to leave a comfortable margin from the k boundary; results are not test-tuned and are stable across the plateau.

Table 14: H^0 subspace dimension m sweep (hard CounterFact retrieval, edge dim $k = 128$). Results are stable across $m \in \{10, 20, 40, 80\}$; $m = 128 = k$ leaves no kernel selection and collapses.

Model	$m=5$	$m=10$	$m=20$	$m=40$	$m=80$	$m=128$	Full
GPT-2	63.7	74.3	85.3	91.2	87.8	35.9	43.1
Phi-2	67.3	75.8	87.1	90.3	85.3	52.8	69.1
Mistral-7B	70.9	82.4	87.1	90.3	86.7	48.5	77.7
Llama-2-7B	62.8	74.0	80.4	85.3	85.1	35.2	68.2

Pooling sensitivity. We compare three pooling choices (mean over all tokens, last non-pad token, first token) by extracting activations under each and rerunning the full H^0 pipeline (Table 15). Mean pooling is best across all four architectures (H^0 at 80–87%); last-token pooling drops 10–30 pp; first-token pooling collapses (autoregressive models have very limited content at position 0). Importantly, sheaf H^0 beats the matched Full hidden state at *every* pooling choice; the decomposition is robust to pooling, but absolute retrieval rates depend on whether the chosen pooling captures content.

Table 15: Pooling sensitivity (hard CounterFact retrieval). Mean pooling is best for all architectures; sheaf H^0 beats Full hidden state under every pooling choice.

Model	Mean (default)		Last token		First token	
	Full	H^0	Full	H^0	Full	H^0
GPT-2	43.1	85.3	25.5	74.3	41.8	42.7
Phi-2	69.1	87.1	26.9	65.9	30.5	43.8
Mistral-7B	77.7	87.1	40.4	72.2	20.5	14.2
Llama-2-7B	68.2	80.4	21.4	47.0	19.6	23.0

C.2 Experimental Details

Compute: All experiments on NVIDIA A100 or RTX 4090 GPUs with 80GB or 24GB VRAM respectively. 7B+ models use 4-bit quantization (bitsandbytes). Sheaf construction (restriction map + Laplacian) takes < 1 minute per model; dominant cost is activation extraction (identical to any activation-based method).

Hyperparameters: Edge dimension $k = 128$ (capturing $\geq 98\%$ of variance). Mean pooling over sequence positions. Restriction map learned via joint PCA. Top/bottom 20 eigenvectors define H^0/H^1 . Steering vectors norm-matched to 30% of mean hidden state norm.

Layer selection: $\ell \approx 2L/3$ (layer 8 for GPT-2, layer 20 for 32-layer models). Targets middle-to-late layers where semantic content is well-formed but output projections have not dominated.

Statistical tests: Bootstrap confidence intervals (1000 resamples). Mann-Whitney U for group comparisons. McNemar for paired binary outcomes. Cohen’s d for effect sizes. All p -values two-sided.

C.3 Practitioner Guidance

Recommended workflow: (1) Build the sheaf with 200+ MRPC paraphrase pairs using PCA restriction maps (parameter-free default). (2) Apply steering at moderate α (0.10–0.15) using both H^1 dims and H^0 -removed, selecting the best on a validation set ($n=50$). (3) For fragile models (identifiable via a single random-steering probe), use sheaf H^1 to avoid generation collapse. For robust models, expect random steering to dominate at $\alpha \geq 0.30$; prefer moderate magnitudes for sheaf advantage.

C.4 NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: Section 1 enumerates four contributions: (i) H^1 functional influence (5.6–26.5 $\times$ over variance-matched controls; Table 2); (ii) H^0 beats LEACE on held-out CounterFact, with a restriction-map ablation showing the kernel selection (not the projection P) drives the result (Table 3, Table 12); (iii) cycles in the paraphrase graph give downstream-useful H^0 (multi-paraphrase rings beat pair-only by +6.5 pp on every architecture; Table 11) and multi-layer matching grids carry $b_1 = N(L - 1)$ algebraic 1-cycles with reproducible Hodge mass (Table 1); (iv) sheaf-regularized SSL prevents collapse (Section 7). Each claim is supported by experiments in Section 5 with effect sizes, p -values, and bootstrap CIs. The honest negative on MLST cycle/transition terms is reported directly in the abstract and Section 7.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 7 contains an explicit Limitations paragraph covering: (a) restriction-map selection is architecture-dependent (contrastive Fisher wins on Llama-2-7B, CCA on Mistral-7B); (b) restriction-map selection is partially domain-dependent (cross-corpus P costs ~ 25 pp, Appendix B.1); (c) sheaf steering wins at $\alpha \in [0.10, 0.15]$ but ties random at $\alpha = 0.30$ on robust models; (d) two node-space operationalizations of relation-hypergraph H^1 did not transfer to retrieval (Appendix B.11); (e) cycle holonomy is architecture-dependent rather than uniformly meaningful (Appendix B.10); (f) the MLST cycle/transition terms are neutral-to-slightly-negative as a training signal (Appendix B.13); (g) our PIXEL comparison uses a simplified port (single global direction; Appendix B.9); we do not empirically compare to Angular Steering or MSRS (orthogonal frameworks).

3. Theory, Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: The paper has one Proposition (Proposition 3.1: reduction of the sheaf Laplacian to projected within-pair covariance). The assumptions are stated explicitly in the proposition body (shared-projection matching-graph regime, one P for every node, $Q_{ij} = I$ on edges); the failure regime under per-node P_v and Procrustes transports is also stated. The full proof is provided in Appendix A.4. Remark 3.2 clarifies the algebraic-vs-spectral H^1 distinction so the cohomological language is unambiguous.

4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

Answer: [Yes]

Justification: The pipeline is specified in equation form in Section 3 (joint-PCA restriction map, sheaf Laplacian as projected within-pair covariance, H^0/H^1 as smallest/largest eigenspaces, mean-difference steering with H^0 -removal). All hyperparameters (edge dimension $k = 128$, H^0/H^1 count $m = 20$, layer $\ell \approx 2L/3$, steering norm $\alpha = 0.3\|\bar{h}\|$, MLP encoder geometry $d \rightarrow 256 \rightarrow 256 \rightarrow 64$, training step counts) are documented in Section 5 and Appendix C.2. Datasets are public (MRPC, PAWS, QQP, STS-B, CounterFact)

with standard splits and 70/30 train/test for the LEACE comparison. Quantization (4-bit bitsandbytes for 7B+ models) and layer selection rules are explicit.

5. Open Access to Data and Code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All datasets used are public benchmarks with standard licenses (cited in Section 5 and below). Code implementing the sheaf decomposition, steering pipeline, multi-layer grid, and MLST will be released on a public repository upon acceptance; we omit the link here for double-blind anonymity. The construction in Section 3 together with the hyperparameters in Appendix C.2 is sufficient to reimplement the pipeline from the paper alone, and all numerical results are deterministic given the published hyperparameters and seeds.

6. Experimental Setting/Details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: Section 5 (Setup and Datasets) and Appendix C.2 specify: nine architectures with sizes; layer selection rule $\ell \approx 2L/3$; edge dimension $k = 128$ chosen to capture $\geq 98\%$ paired variance; H^0/H^1 slice $m = 20$; bootstrap CIs with 1000 resamples; McNemar’s test for paired comparisons; restriction maps (PCA / CCA / contrastive Fisher) and how each is fit; 200 MRPC pairs for sheaf construction; CounterFact $n = 1000$ for steering; 70/30 fact split for LEACE comparison.

7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Bootstrap 95% CIs are reported alongside every steering and retrieval point estimate (Tables 2, 4, 3). McNemar’s exact test is used for paired binary outcomes; Cohen’s d for effect sizes; Mann–Whitney U for unpaired comparisons. Multiple-comparisons control via Benjamini–Hochberg is detailed in Appendix B.8. MLST results are over ten seeds with reported standard deviations and paired t -tests. All p -values are two-sided.

8. Experiments Compute Resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix C.2 reports hardware (NVIDIA A100 80GB or RTX 4090 24GB), 4-bit quantization for 7B+ models, and execution time (sheaf construction < 1 minute per model; activation extraction is the dominant cost). Total project compute is approximately 200 GPU-hours. Section 5 indicates the cluster setup; no experiment requires more than a single A100.

9. Code of Ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: The work is interpretability research on publicly released language models using public paraphrase and factual-recall benchmarks. No human subjects, no crowdsourcing, no scraping of private data, no release of new high-risk artifacts. Datasets used (MRPC, PAWS, QQP, STS-B, CounterFact) are standard licensed benchmarks. We have read the NeurIPS Code of Ethics and conform.

10. Broader Impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The work is interpretability research aimed at improving transparency of large

language model representations. Positive impact: a label-free decomposition that exposes which directions in residual streams are content-stable across paraphrasing and which carry surface variation, potentially aiding red-teaming, alignment audits, and the diagnosis of model fragility (the steering analysis in Section 5.4 is itself a fragility diagnostic). Negative impact: the same machinery could in principle be used to construct targeted style or stance interventions against safety guardrails. We frame steering as a diagnostic tool rather than a deployment mechanism, and we report architecture-dependent failure modes (random steering can outperform sheaf steering at $\alpha = 0.30$ on robust models, which curbs naive misuse).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [NA]

Justification: The paper does not release new pretrained models, generative models, or scraped datasets. We analyze existing publicly released language models (GPT-2, Llama-2/3, Mistral, Qwen, phi-2, Gemma, OLMo) using existing public benchmarks. The released artifact is analysis code, not a model checkpoint or dataset.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All datasets and models are cited at first use: MRPC [15], PAWS [48], QQP [28], CounterFact [34], TruthfulQA [32]. Models are accessed through their official Hugging Face releases under their respective licenses (Apache-2.0 for Mistral and Qwen; the Llama-2/3 community license; MIT for phi-2; Apache-2.0 for OLMo; Gemma terms of use; MIT for GPT-2). Each is used within the terms permitted for academic research.

13. New Assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [NA]

Justification: The paper introduces no new datasets, no new pretrained models, and no new benchmarks. All releasable artifacts are analysis code; documentation will accompany the code release on acceptance.

14. Crowdsourcing and Research with Human Subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [NA]

Justification: The paper involves no crowdsourcing and no human subjects. All evaluation is automated against public benchmarks.

15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [NA]

Justification: No human subjects are involved; no IRB approval is required.

16. Declaration of LLM Usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does not impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [NA]

Justification: The paper analyzes pretrained LLMs (GPT-2, Llama-2/3, Mistral, Qwen, phi-

957 2, Gemma, OLMo) as objects of study; it does not use LLMs as part of its core methodology
958 to generate science content, design the cohomological framework, derive the proposition, or
959 run the experiments. Any LLM-assisted writing or editing is non-substantive and does not
960 change methodology, rigor, or originality.